

Ajay Venkatesh

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EDUCATION

New York University Master of Science in Computer Engineering (GPA – 3.9/4)	Aug 2024 - Aug 2026
Visvesvaraya Technological University Bachelors in ECE (GPA - 8.9/10)	Aug 2017 - Aug 2021

EXPERIENCE

AMAZON, Seattle, US	May 2025 - Aug 2025
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Software Development Engineer Intern

- Designed a secure UI Console architecture using Midway SSO, OAuth, and HELIS to enforce least-privilege, role-based access; presented the solution to the team and received design approval.
- Built serverless infrastructure using AWS CDK to deploy API Gateway, Lambda (Java 21), DynamoDB, and S3 with cross-account access; implemented CI/CD pipeline for multi-stage deployments.
- Developed the frontend using React 18, integrating AWS UI components, Midway AuthN, and HELIS AuthZ with role-aware views and filtering; delivered a UI that was 40% faster and significantly more secure than the previous version.
- Implemented backend APIs using Java Coral, Smithy models, Dagger DI, and CBOR serialization; improved response time by 35% and ensured reliability through comprehensive unit and integration testing.

New York University, NYC, US	Aug 2024 - Present
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Software Developer (On-Campus)

- Contributed to an AI-driven Remote Patient Monitoring System, awarded \$100K by the National Science Foundation, USA, employing React Native, TypeScript, and Cognito authentication for real-time health monitoring.
- Authored parallelized ETL pipelines in Python to convert unstructured NoSQL data into structured SQL using GCP Cloud Functions. Leveraged concurrent processing of data chunks, reducing transformation time by 60%.
- Developed a React website to host QuickSight dashboard handling 50K+ records, utilized Cognito, Amplify, and Lambda functions to establish login, role-based access control, and Stripe integration for payment gateway.

Price Water House Coopers, (PwC)

Senior Software Engineer

Jul 2023 - Aug 2024

- Designed and implemented a distributed backend system on Microsoft Azure for a cloud-based health platform, integrating with Microsoft Dynamics for secure data flow and leveraging Azure services for scalability, fault tolerance, and event-driven processing.
- Led code reviews and mentored 5 interns on software design principles, test-driven development, and performance optimization, reducing production bugs by 25% and accelerating CI/CD pipeline efficiency in an Agile practice.
- Built a scalable B2B web platform with modular backend services to modernize a legacy user onboarding system, integrating data pipelines, email event tracking, and BI report generation, resulting in 70% improvement in automation and extensibility.
- Deployed a cognitive neural model via RESTful APIs to automate structured data extraction from physical cards using OCR and NLP techniques, reducing manual processing time by 75% and enhancing system integration with client applications.

Associate Software Engineer

Aug 2021 - Jun 2023

- Engineered role-based access control (RBAC) and authorization layers, for 5+ different roles, enforcing the principle of least privilege and preventing unauthorized access through strict session validation and privilege escalation checks.
- Optimized a data pipeline by implementing parallel processing for cleaning, mapping, and importing 100K-row Excel files into the database, reducing execution time from 5 hours to 1 hour and improving overall efficiency by 80%.
- Engineered a trie-based autocomplete module to enable fast and dynamic search across users and documents, reducing average query time by 70% and improving operational efficiency for over 10K+ client records.
- Composed client and server-side custom configurations using JavaScript, plugins, custom workflows, webhooks, Business rules yielding a 35% faster server response.

PROJECTS

Advanced Expense Tracker (Java Desktop Application) (Java, JavaFX, JDBC, SQLite, JFreeChart)

- Architected a modular Java application supporting both individual and group expenses with secure login, leveraging OOP principles and design patterns (MVC and DAO) for separation of concerns.
- Implemented concurrent transaction handling and real-time cost splitting using multithreading and synchronized blocks to ensure thread-safe database updates in group expense scenarios.

SKILLS

Programming languages:, Python, Java, C++, JavaScript, SQL.
Web Development: Rest API, GraphQL API, Microservices, React, Node.js, Express.js, Sequelize.
Database: RDBMS (MySQL, PostgreSQL), NoSQL (MongoDB, JSON), Dataverse.
Cloud: AWS (AWS CDK, AWS SDK, EC2, S3, Dynamo DB, API Gateway, Lambda, Cognito, IAM, Q, SageMaker). Azure (AI services, Logic Apps, Security Data lakes, Function Apps, Storage blobs)
Tools: Git, Visual Studio, Jupyter, Postman, Docker, Power Automate, Power BI, QuickSight.
Methodology: Agile, Waterfall, Scrum.
AI/ML: Classification, Pattern Recognition, Data Extraction, MLOps, Neural Network, PCA, LLM.

CERTIFICATIONS

AWS Cloud Practitioner	Microsoft Developer Associate	Power BI Data Analyst Associate
Microsoft Solution Architect Expert	Azure Data Scientist Associate	Azure Data Fundamentals

PUBLICATIONS

Supervised Machine Learning System Based Segmentation and Classification of Stroke Using Deep Learning Techniques ([IEEE](#))
An Aquila-optimized SVM classifier for Diabetes prediction ([IEEE](#))